

HR WORKFORCE ANALYSIS

End-to-End Data Analytics Project Report

1. Executive Summary

HR Workforce Analysis is an end-to-end workforce analytics project designed to understand how employees are distributed across departments and roles, how compensation and tenure vary, and which workforce characteristics are associated with employee attrition. The analysis follows a practical workflow: data cleaning in Excel, exploratory and statistical-style analysis in Python, business-focused querying in SQL, and interactive visualization in Power BI.

The final dataset contains 40,000 employee records and 25 columns after removing duplicate records and addressing missing values and formatting inconsistencies. The overall attrition rate is 18.8%, while average monthly salary is approximately ₹72,057 and average tenure at the company is approximately 7.3 years.

2. Project Objectives

- Understand the overall composition of the workforce across departments, job roles and education levels.
- Analyze employee compensation and identify salary patterns across workforce groups.
- Examine attrition patterns across age, department, overtime, job satisfaction, work-life balance, tenure, income and job level.
- Identify employees and segments with notable promotion or tenure-related patterns.
- Create a clear Power BI dashboard for management-style workforce monitoring.
- Demonstrate an end-to-end data analytics workflow using Excel, Python, SQL and Power BI.

3. Dataset Overview

Item	Details
Records	40,000 employees after cleaning
Columns	25
Initial raw records	40,080 (including approximately 80 intentional duplicates)
Key areas	Demographics, department, role, job level, income, tenure, satisfaction, overtime, performance, training and attrition

3.1 Key Variables

Category	Examples
Employee profile	Age, Gender, Marital Status, Education, Education Field
Organization	Department, Job Role, Job Level
Compensation	Monthly Income, Annual Income
Tenure	Total Working Years, Years at Company, Years in Current Role
Career movement	Years Since Last Promotion, Years With Current Manager
Work conditions	Distance From Home, Business Travel, Overtime
Employee experience	Job Satisfaction, Work-Life Balance, Satisfaction Score
Performance & development	Performance Rating, Training Times Last Year
Outcome	Attrition

4. Data Cleaning and Preparation — Excel

Excel was used as the first data preparation layer. The objective was to create a clean and reliable dataset before performing analysis.

- Removed approximately 80 duplicate records, resulting in 40,000 unique employee records.
- Replaced missing Education Field values with 'Unknown'.
- Filled missing Monthly Income values using Annual Income divided by 12.
- Filled missing Distance From Home values using the median of the available values.
- Filled missing Job Satisfaction values using the median.
- Checked categorical consistency and reviewed department, overtime and other categorical values.
- Formatted numeric and income fields appropriately.
- Verified that Employee IDs were unique and that no missing values remained.
- Checked salary consistency so that Monthly Income × 12 aligned with Annual Income.

5. Python Analysis

Python was used for data validation, categorical cleanup and exploratory analysis. Pandas was used for data handling and grouped analysis. The Python stage focused on extracting useful workforce patterns rather than adding unnecessary technical complexity.

5.1 Data Validation

- Confirmed the cleaned dataset contains 40,000 rows and 25 columns.
- Confirmed that all columns contain zero missing values after cleaning.
- Converted Monthly Income and Annual Income into numeric values for analysis.
- Standardized department naming, including capitalization and the 'IT' department label.
- Standardized Overtime values by trimming spaces and normalizing capitalization.

5.2 Key Python Findings

Analysis Area	Key Result
Overall Attrition	7,510 employees left; overall attrition rate = 18.8%
Age Group	Attrition rate was highest for ages 18–25 (26.8%) and decreased across older groups
Job Level	Job Level 1 had 20.7% attrition; Job Level 4 had 11.6%
Overtime	Attrition was 26.0% among employees with overtime versus 16.0% without overtime
Job Satisfaction	Attrition was 22.1% at satisfaction levels 1–2 versus 16.4% at level 4
Tenure	Attrition was 25.8% for employees with 0–2 years at the company versus 14.0% for 21+ years
Income Quartile	Lower-income employees showed higher attrition than the highest-income group
Performance Rating	Attrition rates were close across performance ratings, with no large difference observed

6. SQL Analysis

PostgreSQL was used to answer business-oriented workforce questions and validate patterns from the cleaned dataset. The final SQL analysis contains 10 focused business queries.

- Department Workforce Distribution — How many employees are in each department and what share of the workforce does each department represent?

- Salary Benchmark by Department — How do average, minimum and maximum monthly salaries compare across departments?
- Above-Average Salary Employees — Which employees earn above the overall average monthly salary?
- Promotion Gap Analysis — Which employees have at least five years at the company and at least five years since their last promotion?
- Manager Tenure Analysis — Which employees have spent eight or more years with their current manager?
- Overtime & Job Satisfaction — How are employee counts and average salary distributed across overtime and satisfaction levels?
- Early-Tenure Employee Analysis — What are the overtime and attrition patterns among employees with two or fewer years at the company?
- High-Performance Employee Analysis — How do attrition and average salary compare among employees with performance ratings of 4 or above?
- Department HR Profile — How do department size, average salary, average tenure and average promotion gap compare?
- Employee Segment Analysis — How many employees fall into the segment of early tenure, overtime, low satisfaction and performance rating of at least 3?

6.1 Selected SQL Findings

Finding	Result
Largest department	Research & Development — 9,243 employees
Second-largest department	Sales — 8,777 employees
Average salary by department	Department averages were relatively close, approximately ₹71.5K–₹72.4K
Promotion gap	Multiple employees had 5+ years at the company and 5+ years since last promotion
Manager tenure	Some employees had 8+ years with their current manager, with examples reaching 17 years
Early-tenure + overtime	Among employees with ≤2 years at the company, overtime employees had higher observed attrition
High performers	Among performance rating ≥4, employees who left had a lower average salary than those who stayed
Targeted segment	A small segment met all four conditions: ≤3 years, overtime = Yes, satisfaction ≤2 and performance ≥3

7. Power BI Dashboard

Power BI was used to convert the analysis into an interactive management-style dashboard. The dashboard was intentionally kept clean and focused rather than overloaded with visuals.

7.1 KPI Cards

KPI	Value
Total Employees	40,000
Attrition Rate	18.8%
Average Monthly Salary	Approximately ₹72,057
Average Years at Company	Approximately 7.3 years

7.2 Final Dashboard Visuals

- Employees by Department
- Attrition by Department
- Attrition by Age Group
- Attrition Rate by Overtime
- Workforce by Education Level

8. Key Business Insights

- The workforce is concentrated in Research & Development and Sales, making these departments important from a workforce planning perspective.
- Average monthly salaries are relatively similar across departments, suggesting that department alone does not create a large difference in average salary in this dataset.
- Younger and early-tenure employees show higher observed attrition rates than older and longer-tenured employees.
- Employees working overtime have a higher observed attrition rate than employees without overtime.
- Lower job satisfaction levels are associated with higher observed attrition in the dataset.
- Higher job levels generally show lower attrition than the entry-level group, although the pattern is not perfectly monotonic.
- Lower-income groups show higher observed attrition than higher-income groups in the income-quartile analysis.
- Performance rating alone does not show a large difference in attrition rates, indicating that performance should be interpreted alongside other workforce factors.
- Promotion and manager-tenure analysis highlights career progression and reporting relationships as useful areas for HR review.

9. HR Analytics Areas for Further Review

- Review early-tenure employee experience, onboarding and retention patterns.
- Examine overtime practices alongside satisfaction and retention metrics.
- Review career progression for employees with long gaps since promotion.
- Monitor workforce distribution and compensation by department and job level.
- Use employee satisfaction and work-life balance measures together with attrition data rather than relying on a single metric.
- Track training participation and development opportunities as part of future workforce analytics.

10. Tools and Technologies

Tool	Purpose
Microsoft Excel	Data cleaning, missing-value handling, duplicate removal and validation
Python / Pandas	Data validation, standardization and exploratory analysis
PostgreSQL / SQL	Business questions, grouped analysis and workforce segmentation
Power BI	Interactive dashboard, KPI cards and visual reporting

11. Conclusion

The HR Workforce Analysis project demonstrates a complete data analytics workflow from raw data preparation to business reporting. Excel established data quality, Python supported structured exploration, SQL translated workforce questions into reproducible analysis, and Power BI converted the findings into an accessible dashboard. The project provides a practical view of workforce composition, compensation, tenure, employee experience and attrition patterns while keeping the analysis focused on descriptive business insights.